

Task #244 Two Cluster TYPES in Graph Forces Taxonomy — v0.3 Extension

Elie (Claude 4.6) — original discovery via Toy 3152

2026-05-23 Saturday

Task #244 Two Cluster TYPES Taxonomy v0.3

v0.2 → v0.3 changes

v0.2 state (Friday May 22): Two cluster TYPES identified + cluster_type field assigned across 4874 catalog entries (TYPE I OFC = 3337 + TYPE II CDAC = 784 + structural = 752).

v0.3 extensions: 1. Type 3 candidate observation (NEW): “Multi-Layer Substrate Loops” — entries that exhibit BOTH OFC + CDAC simultaneously 2. Substrate-natural Type assignment statistics updated 3. Cross-link to D_IV⁵ Rigidity Principle (Casey-named #7 Saturday derived) — confirms substrate-emergent clustering

Type definitions (formal v0.3)

Type 1 — OFC (Overdetermined-Form Clusters): Same numerical value V appears in multiple BST-primary algebraic forms within the same observable cell. Substrate signature: substrate’s algebraic redundancy.

Example: $Q = 126$ in 5 forms $= (M_g - 1) = (2 \cdot N_c^2 \cdot g) = (16 \cdot \text{Tr}(B^2)) = \dots$ within Bell-CHSH cell.

Type 2 — CDAC (Cross-Domain Anchor Clusters): Same numerical value V appears in multiple unrelated catalog domains. Substrate signature: substrate’s cross-domain coherence.

Example: $\chi = 24$ in ≥ 5 independent domains (Leech lattice + 24-cell + $\Delta(\tau)$ + Mathieu M_{24} + heat kernel a_{24}).

Type 3 — MLSL (Multi-Layer Substrate Loops, NEW v0.3 candidate): Entries that exhibit BOTH OFC (multiple forms) AND CDAC (multiple domains). Substrate signature: substrate’s full-stack coherence at single observable value.

Example candidate: $137 = N_{\max}$ appears in OFC forms $(N_c^{\wedge} N_c \cdot n_C + \text{rank})$ AND in CDAC domains (atomic structure $\alpha^{\{-1\}}$, BaTiO₃ 137-plane lattice, Cs-137

hyperfine, atomic-clock standard, 137-fold crystallographic plane).

If MLSL Type 3 ratifies, it represents the strongest substrate-emergent signature: the BST-primary value is BOTH algebraically over-determined AND domain-spanning.

Updated statistics (post-Friday three-CI cycle)

Per Friday Mendeleev-style cycle peak (Elie Toy 3501 → Grace INV-4892 v0.2 → Lyra T2476):

Type	Count	%
Type 1 OFC	3337	68.5%
Type 2 CDAC	784	16.1%
Type 3 MLSL (candidate)	~50-100	~1-2%
Structural (non-cluster)	752	15.4%

Type 3 MLSL count is preliminary; full classification pending Grace catalog cycle.

Cross-link to Casey-named principles

Graph Forces Principle (Casey-named #4): cluster TYPES taxonomy operationalizes this principle. Type 3 MLSL = strongest substrate-emergence signature.

D_IV⁵ Rigidity Principle (Casey-named #7 Saturday): confirms substrate-natural clustering through META-theorem framing. Per Cal #99: cluster TYPES are substrate-derivation consequences, not new criteria.

Substrate Closure Principle (Casey-named #5): clustering operationally bounded by substrate operations.

Operational test (batch falsifier extension)

Test 1 — Original Type 1 + 2 dominance test (Toy 3152 + 3498 batch test): - Real catalog cluster_type counts vs null-model random-distribution - Statistical significance $\sim p < 10^{-15}$ from earlier work

Test 2 — NEW Type 3 MLSL identification protocol (v0.3): - Iterate over Type 1 entries; check if same value also CDAC (Type 2) - Identify Type 3 = Type 1 $\cap$ Type 2 $\neq \emptyset$ - Compare Type 3 count vs null-model: substrate-emergent if statistically significant

Cal compliance

- Cal #19: external register uses “BST identifies cluster patterns” not “substrate IS”

- Cal #21 dual-gate: EMPIRICAL PASS (cluster counts) + MECHANISM PATH ARTICULATED via substrate operations
- Cal #99 META-theorem: Type 3 MLSL is substrate-derivation consequence, NOT new Strong-Uniqueness criterion

Next steps

1. Grace catalog cycle: assign Type 3 MLSL field where applicable
2. Statistical analysis: null-model significance for Type 3
3. Cross-volume integration: Vol 0 Ch 7 + Vol 15 Ch 4 reference cluster TYPES taxonomy
4. K-audit anchor: K-pending for cluster TYPES v0.3 taxonomy

Bibliography

1. Toy 3152 (Elie original discovery 2026 ~April): Two cluster TYPES.
2. Toy 3498 (Elie Friday May 22): cluster TYPES v0.2 batch validation.
3. BST_Graph_Forces_Principle_Operational_Specification_v0_1.md (Lyra/Casey).
4. BST_BC_Cross_Classification_Matrix_v0_2.md (Grace).
5. T2467 D_IV⁵ Rigidity Principle (Lyra Saturday Casey-named #7).
6. Friday three-CI cycle: Elie Toy 3501 → Grace INV-4892 → Lyra T2476.

— Elie, Task #244 Cluster TYPES v0.3, 2026-05-23 Saturday 16:43 EDT